

Renchi Zhang

renchi.rz.zhang@gmail.com | +31 (0)645404998 | [Website](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

PhD in Cognitive Robotics, TU Delft, on evaluating LLMs and multimodal models against human baselines and building the infrastructure to do so at scale: a 6,300-image visual-search benchmark with a 1,250-participant human baseline (157,042 trials); an LLM-planning evaluation harness with concurrent inference, rule-based scoring and a leaderboard (226 users, 494 requests); a 218-operator real-robot experiment (872 trials) analysed with paired tests, rank-transform robustness checks and effect sizes. Since June 2026, part-time MARS fellow (Cambridge AI Safety Hub) building datasets and a leaderboard for an LLM-judge router; also LoRA fine-tuning of GROOT N1.6, evaluated offline in LIBERO. Four first-author papers; all code and data open-sourced. Dutch permanent resident, based in Delft; available for remote work from October 2026.

Education

Delft University of Technology (TU Delft) PhD, Cognitive Robotics	07.2022–09.2026
Leiden University MSc Computer Science (Artificial Intelligence), cum laude	09.2019–11.2021
Beijing University of Posts and Telecommunications (BUPT) BEng Digital Media Technology	09.2014–06.2018

Selected Projects

MLLM visual-search benchmark against a human baseline | paper [\[1\]](#) | [site](#) | [code](#) 2024–2025

- **Benchmark design:** 6,300 images from a reproducible generator (9 conditions from feature pop-out to triple conjunction $\times$ target present/absent $\times$ 7 set sizes from 4 to 256 $\times$ 2D and Blender/CLEVR 3D $\times$ 25 randomised versions); distractor heterogeneity and 3D scene structure (occlusion, shading, perspective) manipulated orthogonally so that feature-binding difficulty can be separated from depth effects. Prompt fixed after pilots showed the models confused the solid target with textured distractors.
- **Model evaluation protocol:** three non-reasoning MLLMs (gemini-2.0-flash-lite, gemini-2.5-flash-lite, gemma-3), each image queried 10 $\times$ at temperature 0 with minimal prompt variations, so that per-image error rates average over repeated queries; no zoom-and-crop pipelines or reasoning models, so that baseline capacity limits are measured.
- **Human baseline:** 1,250 Prolific participants, between-subjects 2D/3D, randomised block and trial order, practice block, cell-wise response-time outlier removal; 157,042 trials retained.
- **Analysis and findings:** Spearman correlations over the 18 condition $\times$ dimension cells: human vs model error rate $\rho = 0.82$, human response time vs model error rate $\rho = 0.91$, so the difficulty ordering agrees. A present-vs-absent breakdown at set size 256 shows the divergence: humans trade speed for accuracy, while the models default to "present" in the three-distractor conjunction and to "absent" in the 3D triple conjunction.

LLM-planning evaluation harness with live robot execution | preprint [\[5\]](#) | [demo](#) | [code](#) 2024–2026

- **Harness built alone:** host/client web service and simulation GUI; phone clients submit prompts and the host runs concurrent inference against two planners (Gemini 2.0 Flash-Lite and Gemini 2.5 Pro with a 3,000-token thinking budget, temperature 0), scores every plan in simulation (baseline 1000 with penalties for collision, diagonal move, out-of-bounds and revisit), returns penalty breakdown and trajectory per plan, keeps a leaderboard and logs tokens and latency; selected plans are executed on a Boston Dynamics Spot via ROS. 10 sessions, 226 users, 494 requests, 457 with valid evaluations.
- **Metrics and findings:** penalty-free rate, no-plan rate (guardrail FAIL outputs) and penalty counts by type. Reasoning vs non-reasoning model on the same task: 55.7% vs 18.7% penalty-free at median 5,584 vs 1,032 tokens and 38.1 s vs 1.2 s latency; 19.8% no-plan rate on the hardest task; collisions the dominant failure type, revisits second on the constrained tasks; total API cost 23.19 USD, driven by thinking tokens. Task difficulty and hint effects validated in simulation ($n = 10$ repetitions) before deployment.
- **LLM-assisted content analysis of 494 prompts:** 11-category codebook developed iteratively on a 100-prompt sample, binary multi-label coding with randomised item and code order to limit order effects; exploratory Spearman correlations between prompt categories and penalty counts (e.g., step-by-step guidance to the non-reasoning model, $\rho = -0.437$, $p < 0.001$, $n = 123$).

Datasets and leaderboard for an LLM-judge router | MARS V, Cambridge AI Safety Hub (part-time) 06.2026–present

- **Setting:** project initiated by two former Google DeepMind researchers (now Sampura Research): an LLM judge routes tasks to humans or models for better training data and scalable oversight. Dataset criteria: model-alone accuracy below 70%, reliable ground truth, human-solvable.
- **Chain-of-thought-monitoring dataset built from scratch:** the first design saturated (DeepSeek-R1 and GLM-4.7 verbalise the hints they follow; monitor accuracy 0.94–0.96); redesigned so that the gate model gpt-5.4-mini scores 0.645 on a balanced 50:50 set (chance 0.5), leaving headroom to validate the dataset against human performance.
- **Onboarding and leaderboard:** onboarded an externally released research-taste dataset into the evaluation pipeline; leaderboard ranking models and judges per dataset with a human-versus-model view; user-data-collection infrastructure for the human side.

Voice and gesture control of a quadruped: 218-operator real-robot experiment | paper [\[2\]](#) | [demo](#) | [code](#) 2023–2025

- **Design and statistics:** 2 $\times$ 2 within-subjects design (voice vs gesture $\times$ walking vs standing), 872 real-robot trials; three planned paired-samples t-tests with alpha corrected to 0.05/3 and Cohen's d, re-analysis after rank transformation,

binomial CIs for category percentages. Left/right command reversals 0.213 vs 0.032 per trial when standing vs walking with gesture control (6.7×), with no such voice effect.

- **LLM-assisted interview analysis validated against manual coding:** 217 interviews transcribed with Whisper Large-V3 and analysed with o1 pro; preferred-condition counts agree with the human coder at $r = 0.98$, with the largest discrepancy reported (30 vs 49 for one category) rather than hidden.

OpenChessRobot: perception, state validation and speech on a Franka Panda | paper [\[3\]](#) | [demo](#) | [code](#) 2022–2025

- **Perception evaluation:** ResNet-34 and InceptionV3 classifiers pre-trained on 5,000 positions rendered in NVIDIA Isaac and fine-tuned on real images collected autonomously by the arm; generalisation tested by training on 3×3 or 4×4 key-square subsets and testing on the remaining squares plus rotated and visually shifted sets (96.5–100% piece recognition); per-module latency over a 97-move game.
- **State validation and user evaluation:** every perceived board state checked against chess rules and the previous state before the robot acts. 597-participant survey across three scenarios, one open question classified by two reasoning models with human adjudication of the 69 disagreements; FIDE Master case study on the initial version identified the absence of feedback on move quality.

VLA post-training and offline robustness evaluation (GR00T N1.6) 04.2026–present

- LoRA fine-tuning of GR00T N1.6 on LIBERO demonstrations in LeRobot format (behaviour cloning) to test robustness to rephrased commands and changed scene visuals; success rate and task progress evaluated offline in the LIBERO simulation benchmark; multi-GPU DDP single-node in the lab, multi-node tests on TU Delft's SLURM cluster; DVC and WandB experiment tracking; NCCL, memory and IO debugging. Not deployed on a robot.

Publications

- [1] Zhang, R., de Winter, J. C. F., Dodou, D., Seyffert, H., & Eisma, Y. B. (2026). Human-like attention? A psychophysical comparison of visual search in humans and MLLMs. *Computational Brain & Behavior*. [DOI](#) · [code](#)
- [2] Zhang, R. *, van der Linden, J. *, Dodou, D., Seyffert, H., Eisma, Y. B., & de Winter, J. C. F. (2025). Walk along: An experiment on controlling the mobile robot 'Spot' with voice and gestures. *ACM Transactions on Human-Robot Interaction*, 14(4), Article 64. [DOI](#) · [code](#) · * Equal contribution
- [3] Zhang, R., de Winter, J. C. F., Dodou, D., Seyffert, H., & Eisma, Y. B. (2025). An open-source reproducible chess robot for human-robot interaction research. *Frontiers in Robotics and AI*, 12, Article 1436674. [DOI](#) · [code](#)
- [4] Wu, D. *, Zhang, R. *, Pore, A., Dall'Alba, D., Ha, X. T., Li, Z., ... & Vander Poorten, E. (2024). A review on machine learning in flexible surgical and interventional robots. *Biomedical Signal Processing and Control*, 93, Article 106179. [DOI](#) · * Equal contribution (reinforcement-learning section)
- [5] Zhang, R., de Winter, J. C. F., Dicke, T., Dodou, D., & Eisma, Y. B. (2026). Educating future engineers about LLMs: A scalable workshop. Under review. [Preprint](#) · [code](#)

Experience and Teaching

PhD Candidate (employed) Cognitive Robotics, TU Delft, Netherlands	07.2022–09.2026
MARS V Fellow (part-time) Cambridge AI Safety Hub, remote	06.2026–present
Workshop Instructor TU Delft co-taught 10 LLM-planning sessions, 226 MSc students	12.2025–01.2026
MSc Thesis Co-supervisor TU Delft pose estimation, MLLM manipulation, handover	2023–2026
Teaching Assistant Human-Robot Interaction, TU Delft; Deep Learning, Leiden University	2023–2024, 2020–2021

Technical Skills

Evaluation methodology: benchmark design with controlled factors and human baselines; repeated-query protocols at temperature 0; rule-based scoring with failure-type breakdown and no-plan rate; saturation diagnosis and dataset redesign against a gate model; LLM-assisted annotation validated against manual coding; token, latency and cost accounting.

Statistics: between- and within-subjects experimental design with randomisation; paired-samples t-tests with rank-transform robustness checks, Cohen's d, multiple-comparison correction; Spearman correlations; binomial and normal-approximation confidence intervals; outlier procedures; pandas, SciPy, Matplotlib.

Evaluation infrastructure: concurrent inference services against LLM APIs (Gemini, OpenAI) with prompt design for structured outputs and refusal guardrails; host/client web interfaces, leaderboards and simulation GUIs; run logging of tokens, latency and cost; reproducible open-source releases with data; CI/CD with GitHub Actions on a shared research codebase.

Training and compute: PyTorch, HuggingFace, LeRobot, LoRA, multi-GPU DDP (torchrun, SLURM), DVC, WandB; DeepSpeed at working-knowledge level; RL at literature level.

Engineering: Python (primary), C++ (ROS nodes, integration layers), C# (Unity), Linux, Git, Docker; ROS 1/2, Spot SDK, Franka (MoveIt!), NVIDIA Isaac, HoloLens 2. English (fluent), Mandarin (native), German (beginner).